

Likun Ni

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Southern University of Science and Technology, Shenzhen, China (518055)

Research interests

Network science, Complex network, Machine Learning, Bibliometrics

Education

2022 – Present **Southern University of Science and Technology** – Shenzhen, China
Major: Statistics, Department of Statistics and Data Science
Undergraduate
Mentors: Yifang Ma, Associate Professors, Department of Statistics and Data Science,
Southern University of Science and Technology
GPA: 3.51/4.00

Honors and scholarships

2023 The Mathematical Contest in Modeling
Successful Participant.

2023 National College Student Statistical Modeling Competition
Provincial Third Prize.

2023 Contemporary Undergraduate Mathematical Contest in Modeling
Provincial Third Prize.

Research experience

January, 2024 – Present(September, 2024) **A research about Concept embedding and authors analysis**
Mentor: Dr. Yifang Ma
This research is mainly used concept embedding method. And we compute every authors' vector. Then analysis their academic domain and academic performance, and find some links between the feature of authors and their academic performance. This Research is in progress.

- July, 2024 – Present(September, 2024) **A research about Use LLM to do entity extraction from news and build a collaboration network**
 Mentor: Dr. Shihui Feng, University of Hong Kong
 This research is mainly used LLM (Large Language Model) method. We use LLM to do entity and relationship extraction from news. The news are mainly about the Covid-19 in Hong Kong. We want to use the entity and relationship to build a collaboration network specially between organizations.
 This Research is in progress.
- November, 2023 – January, 2024 **Analysis of Papers and Their Citations from the Top Four Statistical Journals: through S-field Pentagon Method**
 Mentor: Dr. Yifang Ma (course project)
 This project investigates the papers from four leading statistical journals with a focus on the distribution of articles, citation networks and temporal dynamics.
 For more information: [here](#).
- May, 2024 – June, 2024 **The Small-World Phenomenon: With Restricted Kleinberg's Model Study**
 Mentor: Prof. Yanqing Hu (course project)
 Based on Kleinberg's related work and algorithm model, we added a constraint: long-range contacts can only be generated horizontally or vertically. And then reproduce the Kleinberg's work.
 For more information: [here](#).

Technical skills

Programming languages

Proficient in: Python

Familiar with: Java, R

Software

L^AT_EX, Gephi, ...

Languages

English (fluent), Japanese (JLPT-N2), Chinese (mother tongue)

Others

I am very passionate about scientific research. Despite being an undergraduate student, I am determined to apply for a **Ph.D.** after graduation.

Selected coursework

- *Selected Topics in Frontiers of Statistics I (Graduated course)*: This course introduces some basic theories and algorithms of graphical models and their applications in the Internet, social media, recommendation systems, biology and other fields. This is a course with distinctive interdisciplinary characteristics. Mathematical methods will involve graph theory, generating function, branching process, Bayesian network, factor graph, cavity method, percolation, phase transition, information theory, machine learning, big-data, etc.
- *Advanced Stochastic processes (Graduate course)*: This course offers an introduction to modern stochastic processes, including Markov processes, Poisson processes, Renewal processes, Martingales, Gaussian processes, and Brownian motion. The course will include not only a solid theoretical foundation, but also some applications to statistics and data science, including Queue theory, MCMC, reinforcement learning, and so on.
- *Introduction to Data Science*: This course uses a combination of theory and practice to help students fully understand the basic tools, theories and methods in data science, including: mathematical theories and common methods in data science; analysis and visualization of different data types; complex data cleaning, dimensionality reduction and modeling, etc.
- *Network Science and Computings*: The course aims to guide students to fully understand the emerging science of complex networks, the related methods and their applications. Develop the concept of network science and data science and the capability to solve real problems by utilizing related network analysis software or programming.
- *Time series analysis*: The aim of this course is to present important concepts of time series analysis such as stationarity, stationary time series models (MA, AR, ARMA models), trends and methods for dealing with non-stationarity, nonstationary time series models (ARIMA models, SEASONAL MODELS etc.), parametric estimation, model diagnostic, forecasting, heteroscedasticity time series models, and other selected topics such as Co-integration and Causality, etc.). The course focuses both on the theory of linear time series and on the practical applications with R.
- *Statistical Linear Models*: This course introduces statistical linear models to undergraduate students with basic knowledge in mathematical statistics. The course covers simple linear regression model, multiple linear regression models and other related topics, and it involves the R language for statistical computing.
- *Statistical Computation and Software*: This course aims to guide undergraduate students to carry out statistical analysis in real data problems by utilizing the statistical software R. It helps students deepen their understanding of commonly used distribution functions and reach the goal of solving practical problems using statistical methods and software.